Marc de Lecea

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Summary

Robotics engineer with 4+ years across the full stack — reverse-engineered hardware protocols, built real-time control systems, and deployed production software infrastructure. Increasingly focused on learning-based robotics: supervised- and reinforcement-learning systems for manipulation and decision-making, applied in simulation and on physical hardware. Experienced taking systems from concept to reliable production use.

Experience

Lab Robotics Engineer III, Generate Biomedicines — June 2022 – Present

- Designed and deployed an agentic workflow/worklist designer and platemap transformer with a modular Python API and TypeScript/React frontend, enabling rapid creation of complex plate layouts and automated liquid handler worklists.
- Designed and built two dual-robot-arm workcells from scratch, delivering lights-out scale-up at 1/4 the time and cost of commercial integrators.
- Extended a custom C# Lynx worklist parser with a supervised-learning deck placement system and collision-detection engine — modeling labware, tip, and head-offset geometry to predict preferred layouts and catch invalid configurations before execution — reducing operator error and setup time by ~65%.
- Prototyping a LeRobot-based tube-handling robot, applying imitation learning policies toward autonomous pick-and-place manipulation.
- Built an OpenCV-based computer vision system for deck/labware validation, catching missing or misplaced labware before protocol execution.
- Developed a Python constraints-based scheduling algorithm for in-house robotic workcell orchestration, improving throughput and reducing integration cost.
- Reduced workcell command latency 1000% by redesigning the driver layer around a real-time push-queue architecture.
- Designed an Arduino-based precision dosing system with closed-loop PID control for continuous reagent delivery.
- Built and hosted Dockerized Python APIs on a Kubernetes cluster with real-time Slack alerting and a dashboard for live instrument monitoring, improving system reliability and visibility.
- Mentored junior engineers, fostering technical growth and team collaboration.

Manufacturing Automation Engineer, Hologic Inc. — June 2021 – June 2022

- Reverse-engineered an undocumented protocol for a laser cutter with no vendor API, then built a custom Python driver to dynamically generate G-code and spoof execution commands, all to

replace manual sharpie-written serial numbers with automated engraving — batches of ~120 units in 2 minutes with more legible, durable, higher-capacity codes.
- Prototyped a batch PCB cleaning tool, speeding up flux cleaning by 1500%.

Education

Master of Science in Robotics — May 2022

University of Delaware, Newark, DE
Dual Concentrations in AI and Controls – Cum Laude

Bachelor of Science in Mechanical Engineering — May 2021

University of Delaware, Newark, DE
Minor in Computer Science – Dean's List

Skills and Abilities

Robot Learning: Isaac Sim, LeRobot, reinforcement/supervised/imitation learning
Software & Robotics: Python, ROS2, C, C++, Java, MATLAB, JavaScript, Git, Linux/UNIX
Controls & Embedded: PID control, CAN bus, Kalman filtering (sensor fusion for camera + odometry localization), Arduino/embedded prototyping
Hardware & Fabrication: CAD (Solidworks, Fusion360, OnShape); manual + CNC mill, lathe, laser cutting, FDM/SLA 3D printing
Lab Automation: Programming and servicing Dynamic Devices Lynx, Hamilton Venus, Opentrons, and Tecan Fluent liquid handlers, as well as HighRes Cellario and Retisoft Genera scheduling systems

Personal Projects

Head Coach, FRC Robotics Team 246: Grew program from 4 to ~30 students over 4 years, mentoring hardware/software fundamentals while achieving 9 team awards.
Isaac Sim Legged Locomotion: Built and trained a TARS-inspired simulated legged robot in NVIDIA Isaac Sim, using PPO-based reinforcement learning to develop a stable walking gait from a randomly initialized policy.